

A CANDIDATE PROOF FOR THE ORDINARY CHOW RANK OF THE SIX-BY-SIX PERMANENT

ABSTRACT. Let perm_6 be the permanent of a generic 6 by 6 matrix. We record a conditional project proof candidate for the assertion that, over an algebraically closed field of characteristic zero, its ordinary Chow rank is 32. The upper bound is Glynn's 32-term identity. The lower bound combines a symmetric image-span inequality with a half-defect estimate for quotient derivative symbols of a degree-six Chow term. The finite rational rows used in the half-defect estimate and the final cancellation are recorded explicitly, together with the exact replay boundary. No assertion about border Chow rank or the general n formula is made.

1. STATEMENT AND NOTATION

Let $\mathbb{K}$ be an algebraically closed field of characteristic zero and let

$$V = \text{span}_{\mathbb{K}}\{x_{ij} : 1 \leq i, j \leq 6\}, \quad \dim V = 36.$$

The six-by-six permanent is

$$\text{perm}_6 = \sum_{\sigma \in S_6} \prod_{i=1}^6 x_{i, \sigma(i)} \in \text{Sym}^6(V).$$

Definition 1.1. A degree-six *Chow term* is a product

$$T = \ell_1 \ell_2 \cdots \ell_6, \quad \ell_i \in V.$$

The ordinary Chow rank $R_{\text{Chow}, \mathbb{K}}(f)$ of $f \in \text{Sym}^6(V)$ is the least integer N for which f is a sum of N Chow terms.

For a homogeneous polynomial f , write $\mathcal{D}_q(f) \subseteq \text{Sym}^q(V)$ for the span of all derivatives of f having degree q . Equivalently, $\mathcal{D}_q(f)$ is the image of the appropriate catalecticant. Put

$$E_q = \mathcal{D}_q(\text{perm}_6).$$

The squarefree subpermanents give

$$\dim E_q = \binom{6}{q}^2; \quad \dim E_2 = 225, \quad \dim E_3 = 400.$$

In particular, the middle catalecticant of perm_6 has rank 400.

Theorem 1.2. Assume Proposition 3.1. Then, over $\mathbb{K}$,

$$R_{\text{Chow}, \mathbb{K}}(\text{perm}_6) = 32.$$

The repository does not promote Theorem 1.2 to an unconditional exact-rank result until Proposition 3.1 has a complete independent proof or independently generated exact certificates for every load-bearing local rank. The lower bound rests on that proposition. Its statement must be read literally: it covers arbitrary factor dependencies, repeated factors, and arbitrary quotients of the actual factor span.

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2. A SYMMETRIC IMAGE-SPAN INEQUALITY

Lemma 2.1 (Symmetric image-span lemma). *Let $A_i : W^* \rightarrow W$, $1 \leq i \leq m$, be symmetric linear maps. Set*

$$r_i = \text{rank } A_i, \quad D = \dim \sum_{i=1}^m \text{im } A_i.$$

Then

$$\text{rank} \left(\sum_{i=1}^m A_i \right) \geq 2D - \sum_{i=1}^m r_i.$$

Proof. Choose a surjection $B_i : \mathbb{K}^{r_i} \rightarrow \text{im } A_i \subseteq W$. Symmetry implies $\ker A_i = (\text{im } A_i)^\perp$, so the form induced by A_i on its rank quotient is nondegenerate. Consequently

$$A_i = B_i J_i B_i^*$$

for an invertible symmetric r_i by r_i matrix J_i . Let

$$B = [B_1 \ \cdots \ B_m], \quad J = \text{diag}(J_1, \dots, J_m), \quad R = \sum_i r_i.$$

Then $\sum_i A_i = BJB^*$, while $\text{rank } B = \text{rank } B^* = D$ and $\text{rank}(BJ) = D$. Sylvester's rank inequality gives

$$\text{rank}(BJB^*) \geq \text{rank}(BJ) + \text{rank}(B^*) - R = 2D - R.$$

□

3. THE REQUIRED LOCAL HALF-DEFECT ESTIMATE

Let $T = \ell_1 \cdots \ell_6 \neq 0$ be a Chow term and set

$$L = \text{span}\{\ell_1, \dots, \ell_6\}, \quad U = \mathcal{D}_3(T), \quad u = \dim U,$$

$$F = \mathcal{D}_2(T), \quad R = F \cap E_2.$$

Given a quotient $P : L \twoheadrightarrow D$ of rank d , differentiating cubics in U and then reducing their quadratic derivatives modulo R defines the quotient derivative symbol

$$\beta_{P,R} : U \longrightarrow D \otimes (F/R).$$

Only the rank of this natural map is used below.

Proposition 3.1 (Required half-defect quotient-symbol estimate). *For every nonzero degree-six Chow term T and every quotient $P : L \twoheadrightarrow D$ of rank d ,*

$$(1) \quad \text{rank } \beta_{P,R} + \frac{20 - u}{2} \geq \frac{10}{3}d.$$

Remark 3.2 (Current reduction of the proof obligation). The candidate argument divides the proof by the factor-span dimension $\ell = \dim L$.

For $\ell \leq 4$, a diagonal one-parameter subgroup degenerates T to a monomial in ℓ active variables with all exponents positive. The corresponding middle-catalectic ranks have floors

$$1, \ 2, \ 4, \ 8$$

for $\ell = 1, 2, 3, 4$, respectively. The intersection of the resulting quadratic derivative space with E_2 has dimension at most one. Combining this with the kernel-preimage estimate for intermediate

quotient ranks and the full-quotient injectivity criterion gives the following adjusted rows:

ℓ	$\text{rank } \beta_{P,R} + (20 - u)/2 \text{ for } d = 0, \dots, \ell$
1	(19/2, 21/2)
2	(8, 9, 11)
3	(5, 15/2, 9, 12)
4	(0, 9/2, 8, 10, 14)

The full-quotient assertion uses $E_2^{(1)} = E_3$: a kernel vector would produce a nonzero permanent cubic supported on fewer than nine essential variables, which is impossible.

For $\ell = 6$, the proposed quotient-symbol reduction gives

$$(2) \quad (0, 7, 10, 10, 15, 17, 20), \quad d = 0, \dots, 6.$$

For $\ell = 5$, linear changes of basis reduce the repeated-factor patterns to

$$x_1 x_2 x_3 x_4 x_5 (x_1 + \dots + x_s), \quad 1 \leq s \leq 5.$$

Their proposed adjusted symbol rows are

s	$\text{rank } \beta_{P,R} + (20 - u)/2 \text{ for } d = 0, \dots, 5$
5	(0, 7, 10, 10, 15, 17)
4	(0, 8, 12, 13, 16, 19)
3	(1, 8, 11, 11, 16, 18)
1, 2	(3, 9, 9, 10, 16, 17)

The last row is intended to use the seven-dimensional directional-shadow floor and the one-dimensional kernel gap at quotient rank four. Direct comparison of every entry in all three tables with $10d/3$ proves (1) provided the geometric reductions and the displayed exact symbol rows are established independently. Those derivations are the unresolved proof obligation in this candidate package.

Remark 3.3 (Exact replay boundary). The project replay script checks every rational comparison in the displayed rows and the final cancellation in Section 4. It does not derive the geometric degeneration, the normal-form reduction, the kernel-preimage estimate, or the exact quotient-symbol ranks (2). Those are mathematical inputs to the replay, not consequences of the JSON certificate.

4. THE GLOBAL LOWER BOUND

Assume that

$$(3) \quad \text{perm}_6 = \sum_{i=1}^N T_i$$

with all $T_i \neq 0$. Let

$$A_i = C_{3,3}(T_i) : \text{Sym}^3(V^*) \longrightarrow \text{Sym}^3(V)$$

be the symmetric middle catalecticant and put

$$U_i = \text{im } A_i, \quad u_i = \dim U_i, \quad \delta_i = 20 - u_i, \quad \Delta = \sum_i \delta_i.$$

Set $U = \sum_i U_i$. Since $E_3 \subseteq U$, write

$$h = \dim(U/E_3), \quad \dim U = 400 + h.$$

The sum $\sum_i A_i$ is the middle catalecticant of perm_6 , hence has rank 400. Lemma 2.1 yields

$$400 \geq 2(400 + h) - \sum_i u_i = 800 + 2h - (20N - \Delta).$$

Therefore

$$(4) \quad \boxed{h \leq 10N - 200 - \frac{\Delta}{2}.}$$

We next derive the opposite estimate. Let

$$F = \sum_i \mathcal{D}_2(T_i), \quad Q = F/E_2.$$

Differentiating (3) twice gives $E_2 \subseteq F$. The blockwise quotient derivative symbol is

$$\tilde{\beta} : \bigoplus_i U_i \longrightarrow V \otimes Q.$$

The prolongation identity $E_2^{(1)} = E_3$ implies

$$\ker \tilde{\beta} = \left\{ (g_i) : \sum_i g_i \in E_3 \right\}.$$

Thus $\text{im } \tilde{\beta}$ is naturally isomorphic to U/E_3 , and

$$(5) \quad \text{rank } \tilde{\beta} = h.$$

This identity already accounts for all overlaps among the spaces U_i .

Let L_i be the factor span of T_i . Since $E_2 \subseteq F$ and the first derivatives of E_2 span V , the spaces L_i together span all 36 variables. Choose an arbitrary ordering and define

$$W_i = \sum_{j \leq i} L_j, \quad d_i = \dim(W_i/W_{i-1}).$$

Then

$$(6) \quad \sum_i d_i = 36.$$

Project the first tensor factor of $V \otimes Q$ to $(W_i/W_{i-1}) \otimes Q$. This kills every earlier symbol block. On the current block it is induced by the arbitrary quotient

$$P_i : L_i \longrightarrow L_i/(L_i \cap W_{i-1}), \quad \text{rank } P_i = d_i.$$

Moreover, the natural map

$$\mathcal{D}_2(T_i)/(\mathcal{D}_2(T_i) \cap E_2) \longrightarrow F/E_2 = Q$$

is injective. Hence the projected local ranks add along this filtration. Proposition 3.1, (5), and (6) give

$$(7) \quad h \geq \sum_i \left(\frac{10}{3} d_i - \frac{\delta_i}{2} \right)$$

$$(8) \quad = \frac{10}{3} \cdot 36 - \frac{\Delta}{2}.$$

Therefore

$$(9) \quad \boxed{h \geq 120 - \frac{\Delta}{2}.}$$

Comparing (4) and (9), the complete individual middle-rank defect cancels:

$$120 - \frac{\Delta}{2} \leq 10N - 200 - \frac{\Delta}{2}.$$

It follows that

$$(10) \quad N \geq 32.$$

5. GLYNN'S UPPER BOUND AND CONCLUSION

For a generic n by n matrix $X = (x_{ij})$, Glynn's identity [1] is

$$(11) \quad \text{perm}_n = 2^{1-n} \sum_{\substack{\varepsilon \in \{\pm 1\}^n \\ \varepsilon_1 = 1}} \left(\prod_{i=1}^n \varepsilon_i \right) \prod_{j=1}^n \left(\sum_{i=1}^n \varepsilon_i x_{ij} \right).$$

There are 2^{n-1} summands, and every summand is a product of n linear forms. For $n = 6$, (11) gives

$$R_{\text{Chow}, \mathbb{K}}(\text{perm}_6) \leq 32.$$

Together with (10), this proves the conditional Theorem 1.2. $\square$

6. COMPUTER-ASSISTED COMPONENT

The finite replay uses exact rational arithmetic only. In the project root, the command is

```
python scripts/n6_exact_ordinary_chow_rank_32_candidate.py \
--verify-json data/n6_exact_ordinary_chow_rank_32_candidate.json
```

The line breaks above are typographical; the script path and JSON path are each single arguments. A successful run prints

```
PASS: conditional rank-32 arithmetic payload matches
and verifies:
```

- (1) every displayed half-defect row dominates $10d/3$;
- (2) the lower symbol constant is 120;
- (3) the upper expression is $10N - 200 - \Delta/2$;
- (4) $N = 31$ leaves a gap of 10, while $N = 32$ has gap zero.

As stated in Remark 3.3, this replay certifies the finite arithmetic interface. The unresolved geometric and local-rank arguments remain proof obligations.

Remark 6.1 (Scope). The candidate concerns ordinary Chow rank over characteristic zero. It does not determine border Chow rank. It also does not establish $R_{\text{Chow}, \mathbb{K}}(\text{perm}_n) = 2^{n-1}$ for arbitrary n . Product rank and Chow rank are used synonymously in the relevant literature; see, for example, [2] for the standard product-rank terminology.

REFERENCES

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